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An Amortized Continuous PSF Field for DES Single-Epoch Imaging

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ABSTRACT

Accurate point-spread-function (PSF) models are the dominant systematic in weak-lensing shear measurement. The standard approach fits an independent PSF model per exposure (e.g., PSFEx, PIFF) from that exposure's stars. We instead learn a single *amortized* continuous PSF field: a multi-head attention network takes an exposure's stars as context (position, color, flux, and pixel cutouts) and predicts the PSF at any queried position and resolution, with no per-exposure fitting at test time. The network targets the *pixel-convolved* (effective) PSF—the expected pixel values a point source would produce—rather than a sub-pixel surface-brightness profile, making the directly observable, identifiable quantity the regression target and the exact kernel needed to forward-model galaxies. On 3,301 frozen test exposures of Dark Energy Survey (DES) single-epoch imaging (CCD 31, *grizY*; 73,609 reserved stars), the model matches or exceeds PIFF and PSFEx on every metric: size-residual scatter 0.030 versus 0.039/0.037, ellipticity errors $|\delta e_1|, |\delta e_2| = 0.015/0.015$ versus 0.017/0.016 (PIFF), and shape correlations 0.65/0.61 versus 0.61/0.58 (PIFF), at $\chi^2/\mathrm{dof} = 1.04$. In a star-anchored galaxy injection–recovery test on real data, our PSF recovers galaxy *ellipticity* far better than PIFF ($\Delta e_1 = +0.004$ versus -0.138), the weak-lensing-critical quantity. Color conditioning removes a chromatic PSF-size systematic that the baselines retain ($+0.004$/mag versus ~ -0.06/mag residual slope). The model is flat in the number of available fit stars where PIFF degrades sharply ($\chi^2 \simeq 1.1$ versus 5.1 at five stars). Using simulations as a controlled instrument, we show that a *blend forward-model* training loss removes a size bias from unrecognized blends, and we identify and fix a spin-2 representation failure in which the network corrupts the PSF ellipticity by attending to galaxy detections in its context; restricting attention to point sources recovers it fully.

Keywords: Weak gravitational lensing (1797) — Point spread function (1349) — Astronomical techniques (1684) — Convolutional neural networks (1938) — Sky surveys (1464)

1. INTRODUCTION

Weak gravitational lensing infers the matter distribution from coherent ∼1% distortions of galaxy shapes. The measurement is limited by knowledge of the PSF: any error in the modeled PSF size or ellipticity propagates directly into the inferred shear. Survey pipelines model the PSF per exposure by fitting that exposure's stars with a flexible model interpolated across the focal plane (PSFEx; E. Bertin 2011) (PIFF; M. Jarvis et al. 2021). This is robust but (i) discards information shared across exposures and nights, (ii) degrades when few clean stars are available, and (iii) is purely empirical at the star positions—it does not exploit auxiliary information such as stellar color, which sets the chromatic PSF.

We take an amortized, learning-based approach. A single network is trained across many exposures to map an exposure's stars (the *context*) to a continuous PSF field, queryable at arbitrary position, color, flux, and resolution. The inferential target is the PSF at *star-free* positions, where galaxies are measured; predicting held-out stars is the training objective and the validation instrument. We train and evaluate on DES single-epoch data and benchmark against PIFF and PSFEx run by us under matched conditions.

A second, more fundamental difference is the *target* of the model. Traditional PSF models—including PSFEx (super-sampled pixels) and PIFF (a surface-brightness profile on the sky)—represent the PSF as a continuous function on $\mathbb{R}^2$ at finer-than-pixel resolution, which must then be convolved with an assumed pixel-response kernel to predict what a detector records. We instead model the *effective* PSF (ePSF; J. Anderson & I. R. King 2000): the instrumental PSF already integrated over the pixel, i.e. the expected fraction of a point source's light falling in each pixel as a function of the source's sub-pixel position. The network maps a query position (and color, flux) directly to these expected pixel values, never representing the continuous PSF explicitly—the sense in which our PSF is *implicit*. This target is more principled on two grounds. First, identifiability: from sampled data one cannot uniquely recover the sub-pixel surface brightness without extra assumptions, whereas the ePSF is exactly what the pixels constrain—the reason it underpins state-of-the-art point-source astrometry and photometry (J. Anderson & I. R. King 2000; T. R. Lauer 1999; T. I. Liaudat et al. 2023). Second, sufficiency: the ePSF as a function of source position is a sufficient statistic for forward-modeling *any* source—a galaxy's expected pixel data is its surface brightness convolved with the ePSF, with no separate pixel step (G. M. Bernstein & M. Jarvis 2002)—so the ePSF is precisely the kernel a shear pipeline needs. The only residual is discretizing that convolution on a finite grid, which our ability to render at arbitrary oversampling controls directly.

2. METHOD

Architecture—Each detected source is encoded from its pixel cutout (a learned image encoder applied to the scale-normalized stamp), its sky position (sinusoidal encoding), and scalar color and log-flux. A multi-head self-attention block lets every query attend to the context sources; a coordinate decoder then renders the PSF at the query position and any oversampling factor, returning the pixel-convolved effective PSF used to convolve galaxy models. We use a polar (spin-2-aware) angular parametrization of the decoder and FiLM modulation of the decoder by the attended context features.

Inferential target (effective PSF)—The decoder outputs the effective PSF (J. Anderson & I. R. King 2000)—the pixel-integrated point-source response—sampled at the query grid, optionally oversampled. Because the loss below compares predicted stamps to *observed* star pixels, the model regresses the observable by construction: it learns $\mathbb{E}[\text{pixel value} \mid \text{point source at } (x, y)]$ directly, rather than fitting a sub-pixel profile and assuming a pixel kernel. This is also what makes the benchmark fair: we score all three methods on moments measured in pixel space (model stamp versus star stamp), so no method is privileged by its internal representation. The same pixel-convolved output is the kernel used to forward-model galaxies in §4.3; oversampling the render reduces the discretization error of that convolution for compact sources.

Loss—Training minimizes an inverse-variance χ^2 between predicted and observed star cutouts, with the per-star amplitude solved in closed form. We introduce a *blend forward-model* loss: rather than treat each clean star as isolated, we model its cutout as a generalized-least-squares sum of the target plus its detected neighbors, all rendered by the same decoder (§4.6). Detected galaxies near a target are handled by exclusion (default), masking, or a crude extended-component model; the three are within noise on real reserved stars.

Point-source context—The PSF should be informed only by point sources. We restrict the attention context to stellar detections; §4.6 shows this is necessary to avoid a spin-2 corruption from extended sources in the context.

Color conditioning—The decoder conditions on $g-i$ color, enabling a per-object chromatic PSF (differential chromatic refraction and wavelength-dependent seeing) that per-exposure empirical models cannot represent (§4.4).

3. DATA

We use DES single-epoch FITS (Y5A1, CCD 31; amplifier B is dead, masked via MSK bit 8). Sources are detected with SEP and cross-matched to DES DR2 for colors, `SPREAD_MODEL` star/galaxy labels, and coadd object IDs. Splits are at the observing-night level by deterministic hash to prevent leakage (consecutive exposures share atmosphere), frozen in a committed manifest with $\sim$20% of clean stars per test/validation exposure reserved for scoring only; the test split is 3,301 exposures across $grizY$. Reserved stars are excluded from every method's fit/context and scored identically across methods on the same pixel grid.

Simulations—A matched pipeline renders Moffat PSF fields (second-order polynomial FWHM and shear variation—exactly representable by PIFF/PSFEx, a deliberately fair-to-baselines choice) into the same data format, optionally with chromatic FWHM–color dependence, blended stars, and injected galaxy detections. Star density, galaxy density (3:1), and per-source S/N are tuned to match the real extraction. The simulation is the controlled instrument for the systematics studies in §4.6.

4. RESULTS

4.1. *Reserved-star accuracy on real data*

On 3,301 test exposures and 73,609 reserved stars, the amortized model (no per-exposure fitting) matches or

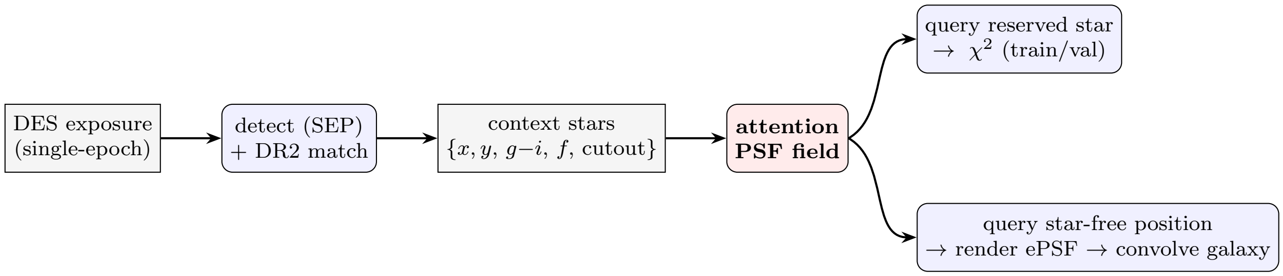

Figure 1. Overall workflow. A single network, trained across exposures, maps an exposure's detected stars (context) to a continuous PSF field. Querying held-out reserved stars provides the training objective and validation; the inferential target is the PSF at star-free positions, rendered (oversampled) to convolve galaxy models.

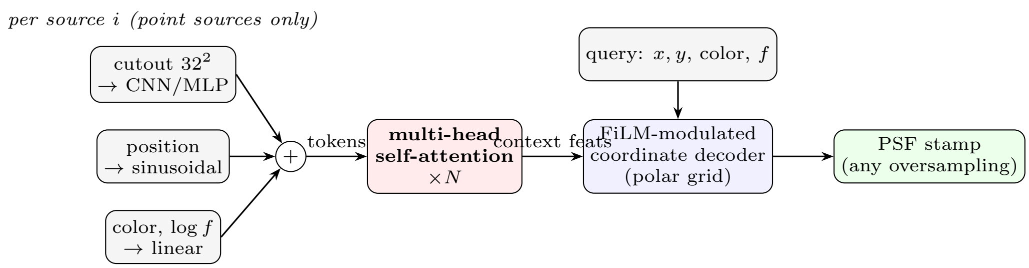

Figure 2. Network architecture. Each context source (restricted to point sources; §2) is encoded from its pixel cutout, sky position, and scalar color/log-flux into a token. Multi-head self-attention mixes the context; a FiLM-modulated coordinate decoder, conditioned on the attended features and the query (position, color, flux), renders the pixel-convolved PSF at any position and oversampling factor.

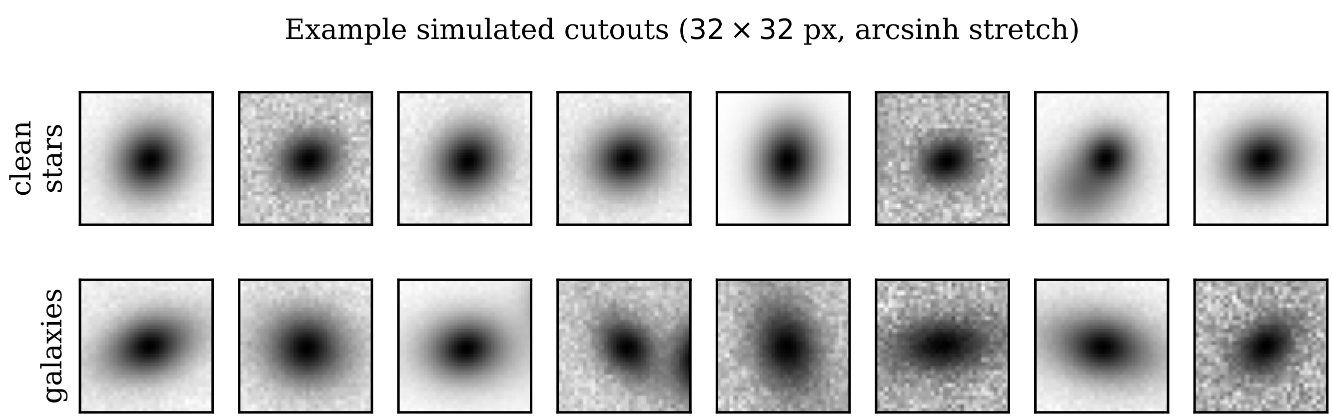

Figure 3. Example simulated cutouts. Clean stars (top) are compact point sources; injected galaxies (bottom) are visibly extended and elliptical (second-moment size $T \approx 33$ vs $15\,\mathrm{px}^2$). Both are presented to the network as context.

exceeds both baselines on every metric (Table 1). Reserved "clean" stars carry a mild blend-contamination floor that suppresses absolute correlations for *all* methods; the comparison is fair (identical stars) and the ordering is robust.

4.2. *Residual two-point statistics*

Per-star residuals understate the impact on weak lensing, where the *spatially coherent* part of the PSF-modeling error sources additive shear bias. We therefore compute the standard ρ-statistics (B. Rowe 2010; M. Jarvis et al. 2016) of the reserved-star residuals—the auto- and cross-correlations of the residual ellipticity δe and the size-leakage term $e\,\delta_T$—accumulated within exposures with TREECORR (Figure 6). On the same reserved stars used for Table 1, the amortized model is at or below both per-exposure baselines on four of the five statistics: the median $|\xi_+|$ across angular scales is 3.9×10^{-5} (ρ_1), 5.2×10^{-5} (ρ_2), 2.5×10^{-6} (ρ_3), and 4.8×10^{-6} (ρ_4), against $\rho_1, \rho_2 \gtrsim 7 \times 10^{-5}$ and $\rho_3 \simeq 6 \times 10^{-5}$ for PIFF. The strong ρ_3 result—more

Table 1. Reserved-star residuals on the DES test split (3,301 exposures, 73,609 stars). T scatter is the robust standard deviation of $\delta T/T$; $|\delta e|$ are median absolute ellipticity residuals; corr is the model–star shape correlation. Best in bold.

Method	T scat	$\lvert\delta e_1\rvert$	$\lvert\delta e_2\rvert$	corr(e_1)	corr(e_2)	χ^2/dof
This work	**0.0301**	**0.0151**	**0.0150**	**0.650**	**0.607**	1.037
PIFF	0.0393	0.0171	0.0158	0.611	0.577	1.073
PSFEx	0.0371	0.0160	0.0154	0.645	0.604	1.047

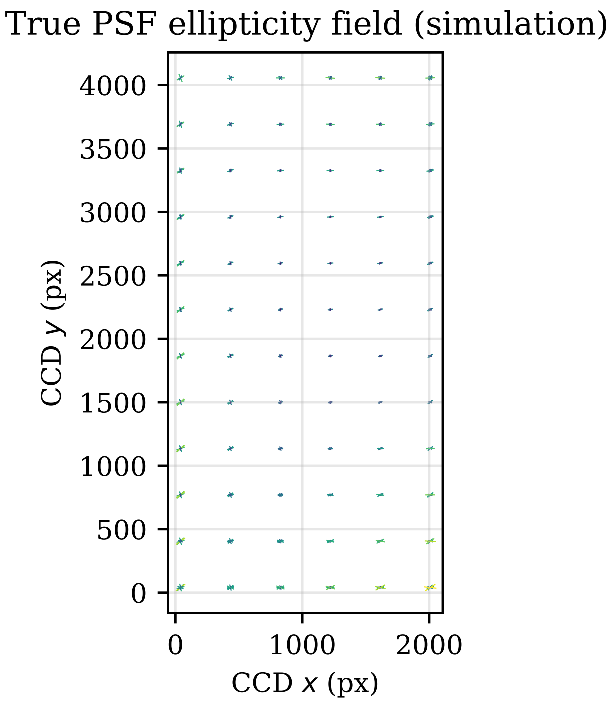

Figure 4. The true, spatially varying PSF ellipticity field in a simulated exposure (whisker length $\propto |e|$); the network must recover this at star-free positions.

than an order of magnitude below PIFF—reflects the model's smaller and less spatially structured size residuals. The one exception is $\rho_5 = \langle e\,(e\,\delta_T)\rangle$, where our model ($5.4\times10^{-5}$) sits above the baselines ($\sim10^{-5}$): a residual coupling between PSF shape and the size error that a weak-lensing analysis would carry in its additive-bias budget. Overall the coherent, lensing-relevant part of the residual is no larger than that of the per-exposure baselines, despite the amortized model fitting no per-exposure free parameters.

4.3. *Galaxy recovery on real data*

We test the inferential goal directly. We inject GALSIM galaxies, convolved with a strictly isolated reserved star's empirical PSF, at that star's position; each

Table 2. Star-anchored galaxy injection–recovery (113 r-band exposures, 522 anchors). The truth arm validates the protocol.

Arm	Size bias	Δe_1	Δe_2
Truth (validation)	−1.5%	−0.002	∼0
This work	−7.6%	+**0.004**	−0.002
PIFF	−1.0%	−**0.138**	−0.001

method predicts the PSF there from the *other* stars and recovers the galaxy. The truth arm (the anchor's own image) recovers near-zero bias, validating the protocol. On 113 r-band exposures and 522 anchors (Table 2), our PSF recovers galaxy *ellipticity* essentially without bias, whereas PIFF's PSF-ellipticity error induces a large shape bias ($\Delta e_1 = -0.14$). PIFF recovers size slightly better; our PSF is $\sim$6% too broad at star-free positions. For weak lensing, the ellipticity result is the consequential one.

4.4. *Chromatic PSF correction*

In a chromatic simulation (FWHM varies −3%/mag of $g-i$), color conditioning reduces the residual chromatic size slope from −0.06/mag (PIFF, PSFEx, and our own zero-color ablation) to +0.004/mag—a $\sim$17× reduction. On real r-band data the paired color-versus-zero-color slope difference is +0.0024/mag (CI [+0.0016, +0.0030]); the baselines structurally retain the systematic. This is information unavailable to a per-exposure empirical PSF.

4.5. *Sample efficiency*

Restricting every method to k randomly chosen fit stars, the amortized model is approximately flat in k (size scatter $0.078 \to 0.056$, $\chi^2 \sim 1.1$ from $k = 5$ to $k = 100$), while PIFF collapses below $k \approx 25$ ($\chi^2 = 5.1$, scatter 0.18 at $k = 5$). Amortization transfers information from the training set to sparsely-populated exposures.

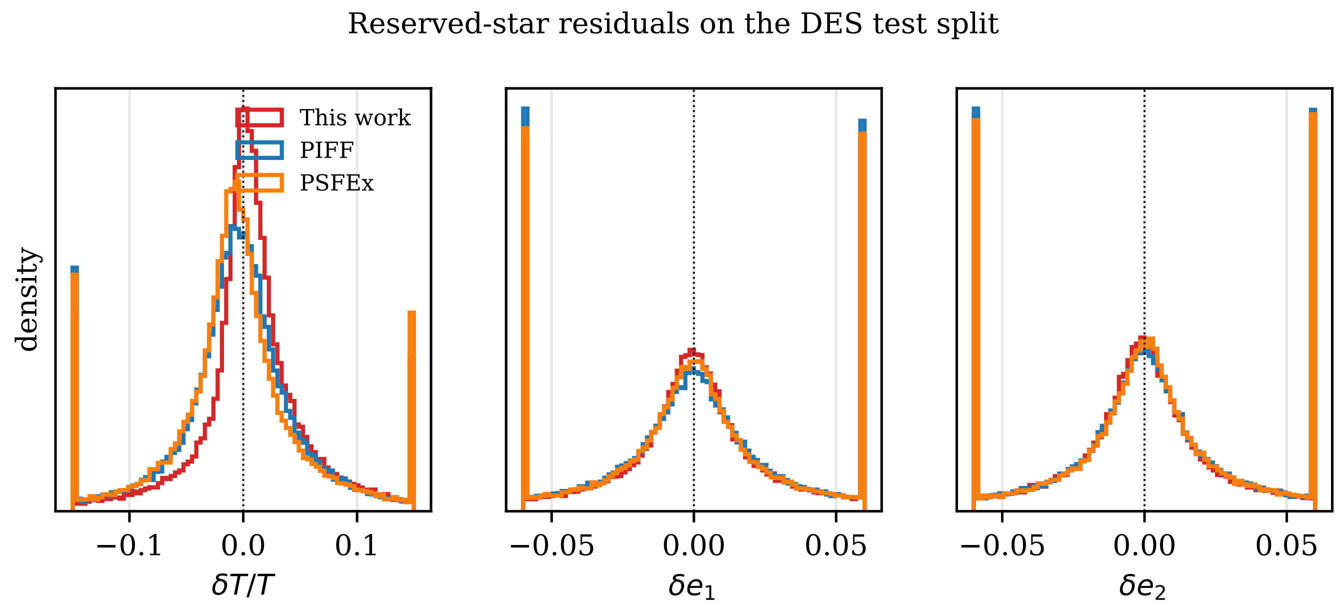

Figure 5. Reserved-star residual distributions on the DES test split. Our amortized model (red) is the narrowest in size and ellipticity.

4.6. *Simulation studies*

Architecture ladder—On the star-free truth grid, χ^2/dof improves along additive → FiLM → diagonal → polar (8.9 → 4.96 → 4.41; PIFF 2.67; verified floor 1.0), recovering both ellipticity components.

Blend forward-model—On a blended simulation, single-star training leaves a +4.8% size bias on the star-free truth grid; the blend forward-model loss reduces it to +0.7% (≈7×).

Galaxy-context corrupts PSF ellipticity—and the fix—Injecting galaxy detections into the simulation, a model that attends to *all* detections (including galaxies) suffers a clean, reproducible failure: it cannot represent one spin-2 ellipticity component [corr(e_1) = 0.05 under the polar parametrization; the failure flips to e_2 under a Cartesian parametrization, identifying a representation degeneracy rather than noise], while recovering the other perfectly. We localize the cause by ablation: it is *not* data volume, per-source S/N, galaxy size or color, or the χ^2 cap—the model is treating extended galaxy cutouts as PSF evidence. Restricting the attention context to point sources recovers *both* components completely [corr(e_1, e_2) = 0.98, matching the galaxy-free ideal]; a coordinate-system change does not. Galaxies are clearly distinguishable (cutout second moment $T \approx 33$ versus $15\,\mathrm{px}^2$ for stars), so this is the model failing to use available information, fixed by the principled restriction that only point sources inform a PSF. On the real test set this restriction carries no accuracy cost: a point-source-context model matches the headline model's reserved-star residuals (size scatter 0.031 versus 0.030; identical median ellipticity error and shape correlation), confirming that the fix generalizes from simulation to data without trading away real-world performance.

5. DISCUSSION

The amortized field trades a per-exposure fit for a single trained model. Benefits: no test-time fitting, robustness in star-sparse exposures, conditioning on color (chromatic PSF) and flux (brighter–fatter), and a directly forward-modelable output—we predict the pixel-convolved effective PSF, the identifiable quantity the data constrain and the exact kernel a galaxy model is convolved with (§1), so the PSF feeds a shear pipeline with no intervening pixel-deconvolution assumption. Caveats: (i) unlike per-exposure fitters, the amortized model requires many exposures to learn the field (in simulation, ellipticity recovery is unstable below a few thousand exposures; real DES is well above this); (ii) the simulation's idealized PSF (Moffat, low-order polynomial field) favors the per-exposure baselines, so absolute-accuracy comparisons rest on real data; and (iii) the galaxy-context finding argues that a PSF estimator's context should be gated to point sources, which we adopt. (iv) A residual *magnitude-dependent* size bias remains: the median $\delta T/T$ rises to $\sim$+1.2% at intermediate magnitudes (Fig. 13), where the per-exposure baselines stay within ±0.5%; this is consistent with the $\sim$6% over-broadening in galaxy recovery (§4.3) and in-

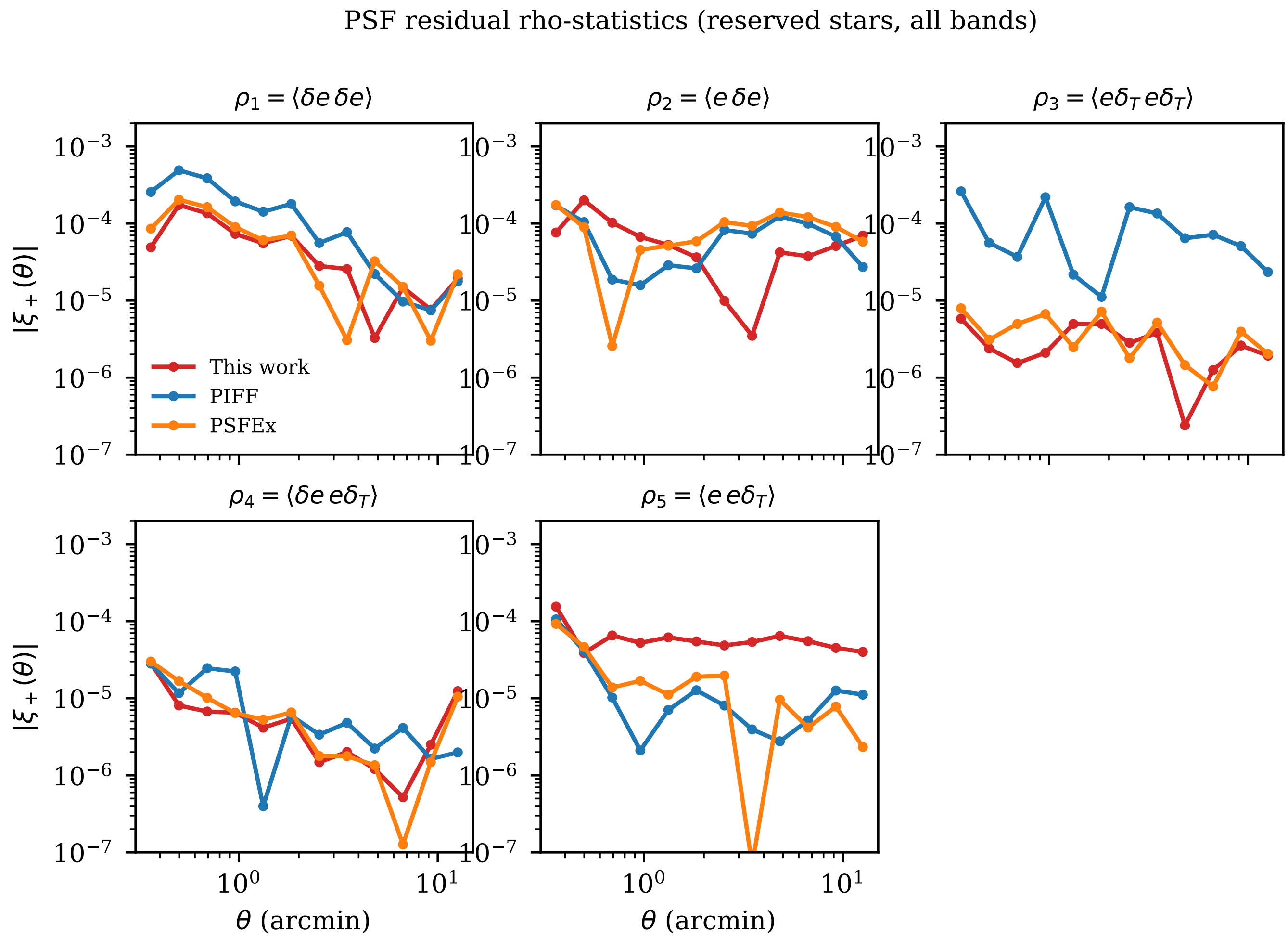

Figure 6. PSF residual ρ-statistics on reserved stars, per method. Our model (red) is at or below both baselines on ρ_1–ρ_4; only ρ_5 (the shape–size coupling) is modestly larger (§4.2).

dicates that log-flux conditioning only partially captures brighter–fatter—the main residual systematic to address before a shear analysis.

6. CONCLUSION

A single attention-based continuous PSF field, trained across DES single-epoch exposures and applied with no per-exposure fitting, matches or exceeds PIFF and PSFEx on real reserved stars and recovers galaxy ellipticity better than PIFF on a star-anchored injection test, while additionally correcting a chromatic systematic and remaining robust in star-sparse exposures. Controlled simulations motivate two training choices—a blend forward-model loss and point-source-only attention context—the latter from a clean diagnosis of a spin-2 representation failure triggered by galaxy detections in context.

ACKNOWLEDGMENTS, FUNDING, DES CONSORTIUM STATEMENT.

Baselines: PIFF (PixelGrid, second-order `BasisPolynomial`, 4σ rejection) and PSFEx (`PSF_SAMPLING` 0.5, 45×45, `PSFVAR_DEGREES` 2; rendered via GALSIM `DES_PSFEx`, `method='no_pixel'`).

Software: Astropy (Astropy Collaboration 2022), GalSim (B. T. P. Rowe et al. 2015), SEP (K. Barbary 2016), PyTorch (A. Paszke et al. 2019), PIFF (M. Jarvis et al. 2021), PSFEx (E. Bertin 2011)

APPENDIX

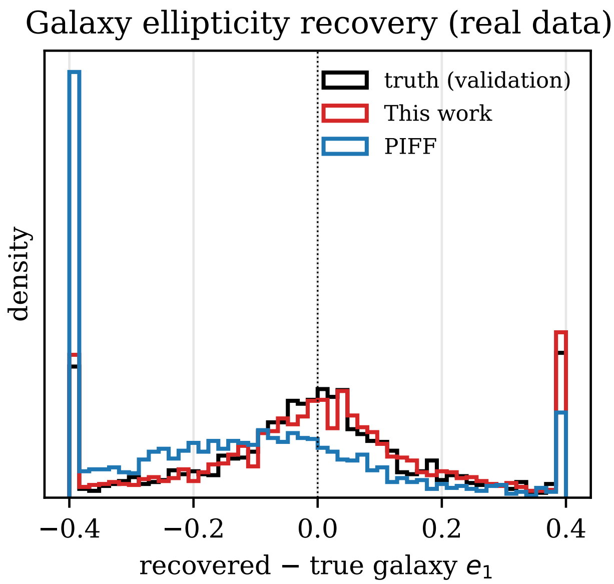

Figure 7. Galaxy ellipticity recovery on real data (star-anchored injection). PIFF's PSF-ellipticity error biases recovered galaxy e_1 by -0.14; ours is unbiased.

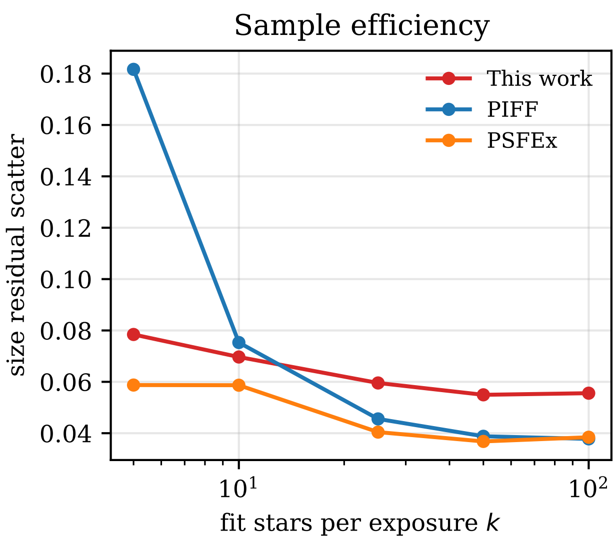

Figure 8. Size-residual scatter versus the number of fit stars k. The amortized model is nearly flat in k; PIFF degrades sharply below $k \approx 25$.

A. REAL-DATA STAR SELECTION AND PSF EXAMPLES

Figure 10 shows a region of a real DES single-epoch r-band exposure with the source selection overlaid: clean stars used for fitting and scoring (green), held-out reserved stars (blue), and additional context-only stellar detections (orange). Galaxies (the extended sources, e.g. the bright object near the center) are detected but, under the point-source-context policy (§2), are not attended to. Figure 11 compares, for a single high-S/N reserved test star, the observed stamp against the PSFEx, PIFF, and ImplicitPSF models evaluated at that position (each method having seen only the non-reserved stars). All three reproduce the stellar profile; PIFF's per-exposure interpolation is visibly noisier in the wings.

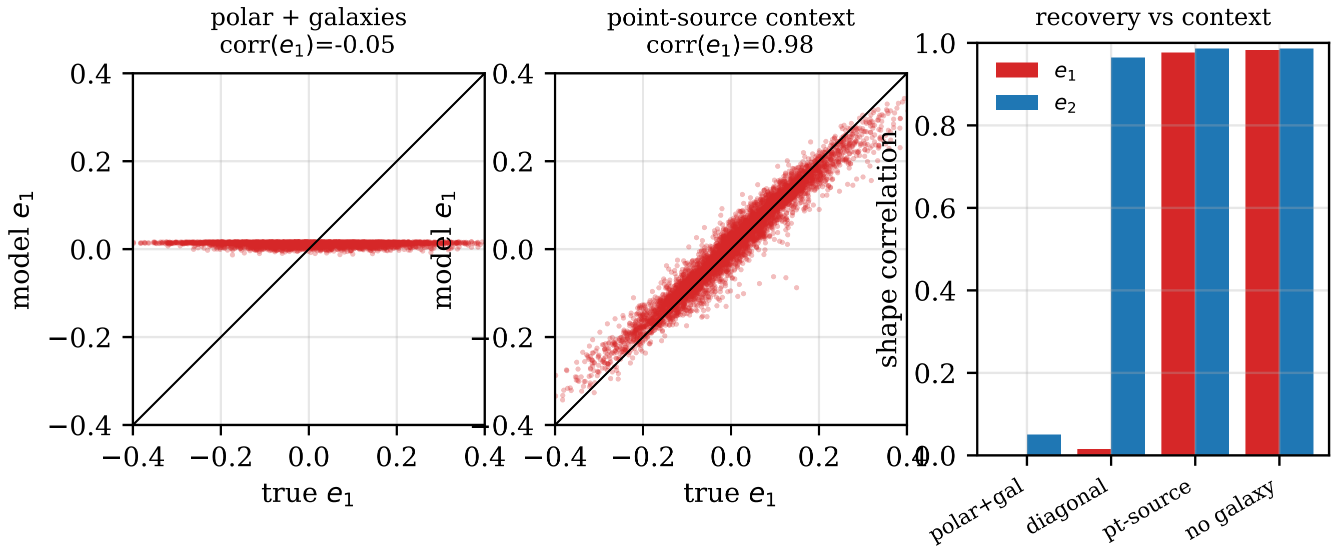

Figure 9. Galaxy detections in the attention context corrupt the recovered PSF ellipticity. Left: with galaxies in context, model e_1 collapses to a constant (corr$= -0.04$). Middle: restricting the context to point sources recovers it (corr$= 0.98$). Right: shape correlations across configurations—point-source context and galaxy-free both recover *both* components; a coordinate change (diagonal) does not.

B. BENCHMARK VALIDATION DIAGNOSTICS

For comparison with standard per-exposure PSF validation (cf. M. Jarvis et al. 2021), we report the same diagnostics. Table 3 breaks the headline metrics down by band. The amortized model has the smallest size scatter and median ellipticity error in every band, with the largest margins in Y and z, where the broader, more variable PSF is hardest for per-exposure fits.

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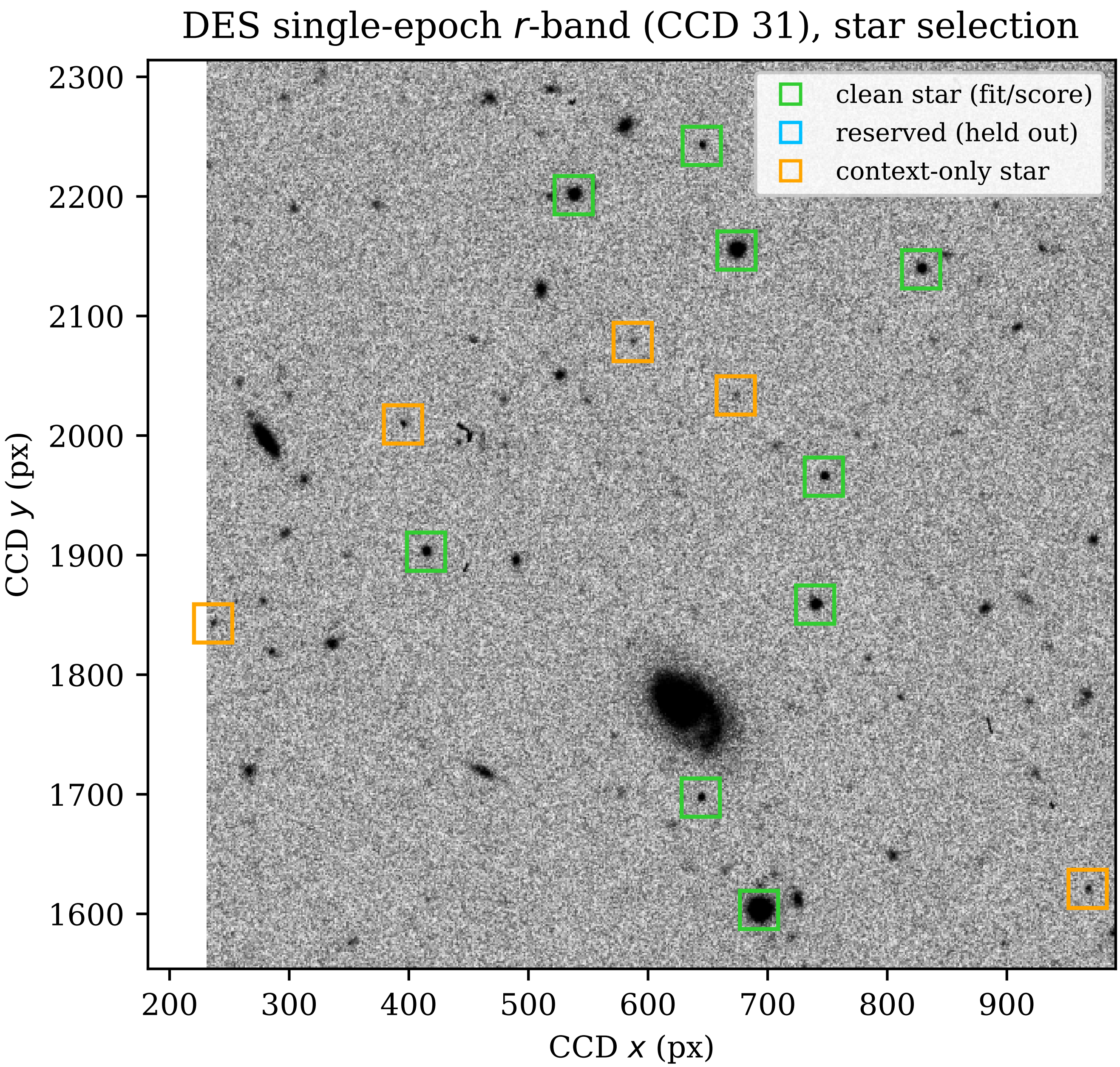

Figure 10. Star selection on a real DES exposure. Green: clean stars (fit and score). Blue: reserved (held out for evaluation). Orange: context-only stellar detections. Extended sources (galaxies) are excluded from the attention context.

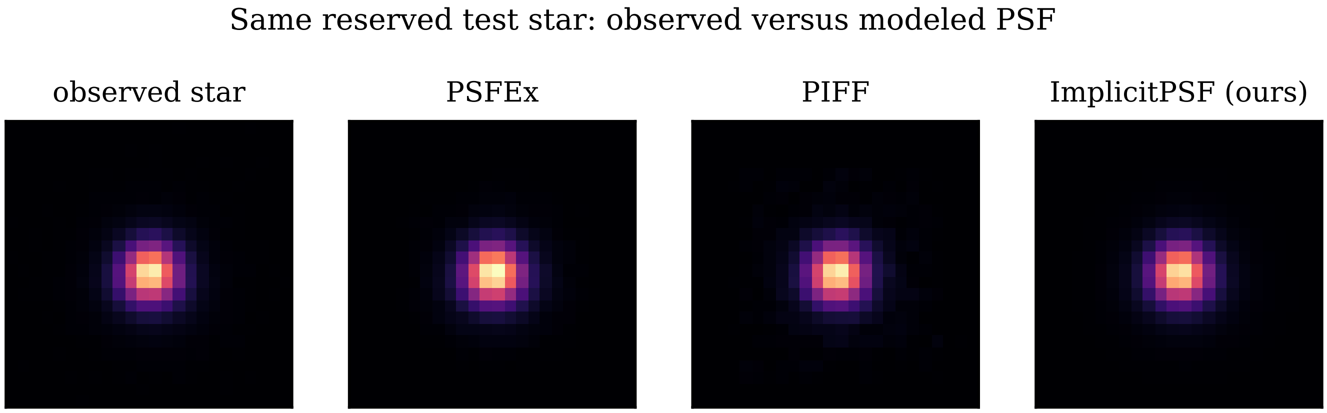

Figure 11. The same reserved test star (left, observed) and the PSF predicted at its position by PSFEx, PIFF, and ImplicitPSF. Each model was fit/conditioned on the non-reserved stars only. Common linear stretch.

Table 3. Reserved-star residuals by band (test split). T scatter is the robust standard deviation of $\delta T/T$; $|\delta e_1|$ is the median absolute e_1 residual; corr is the model–star e_1 correlation. Best per band in bold.

Band	Method	T scat	$\|\delta e_1\|$	corr(e_1)
g	**This work**	**0.0331**	**0.0164**	**0.475**
	PIFF	0.0439	0.0188	0.448
	PSFEx	0.0417	0.0173	0.473
r	**This work**	**0.0291**	**0.0144**	**0.576**
	PIFF	0.0366	0.0164	0.535
	PSFEx	0.0340	0.0151	0.572
i	**This work**	**0.0274**	**0.0136**	**0.643**
	PIFF	0.0342	0.0155	0.605
	PSFEx	0.0339	0.0144	0.637
z	**This work**	**0.0272**	**0.0140**	**0.601**
	PIFF	0.0387	0.0160	0.503
	PSFEx	0.0359	0.0152	0.589
Y	**This work**	**0.0391**	**0.0210**	**0.568**
	PIFF	0.0513	0.0226	0.523
	PSFEx	0.0449	0.0221	0.550

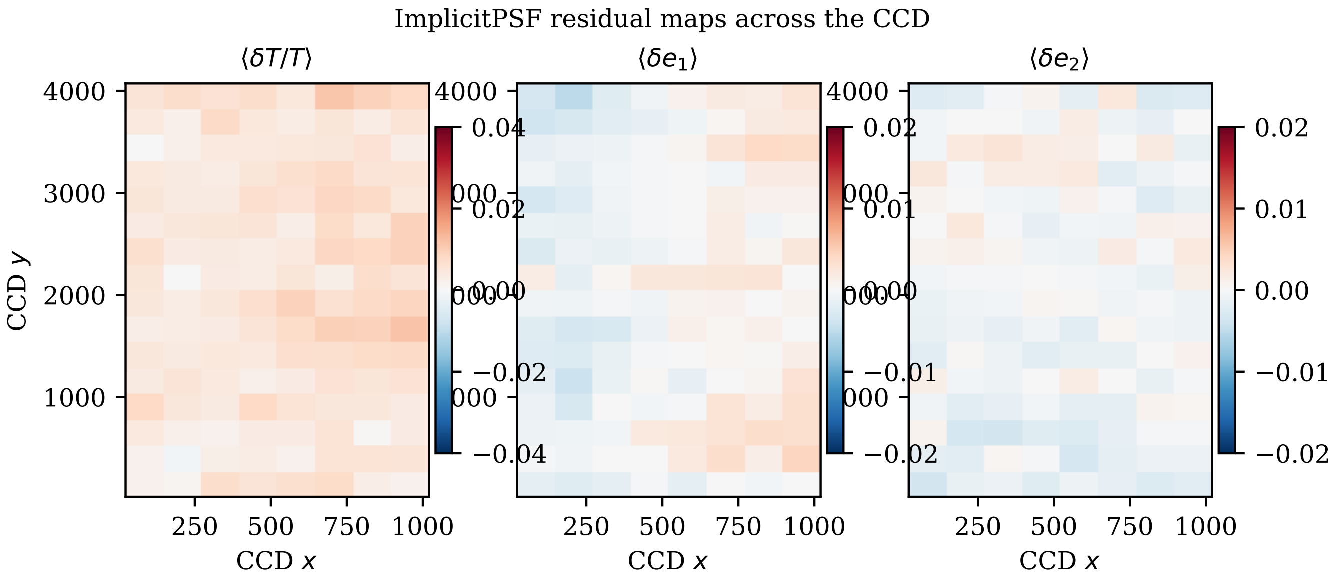

Figure 12. Median ImplicitPSF residuals binned across the CCD (cf. PIFF Fig. 9).

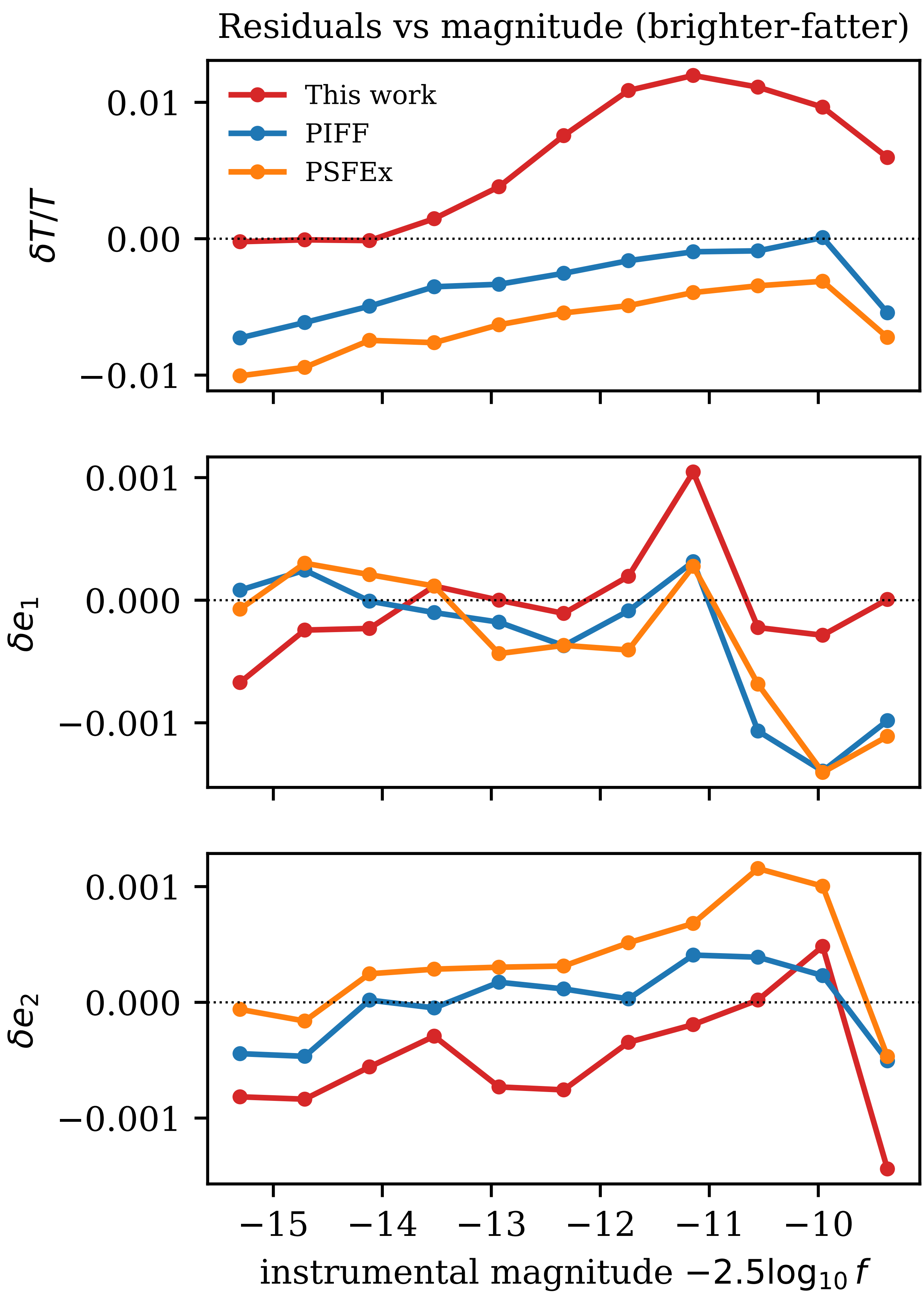

Figure 13. Residuals versus instrumental magnitude (cf. PIFF Fig. 10). Our model (red) carries a magnitude-dependent size residual peaking at +1.2% (§5).

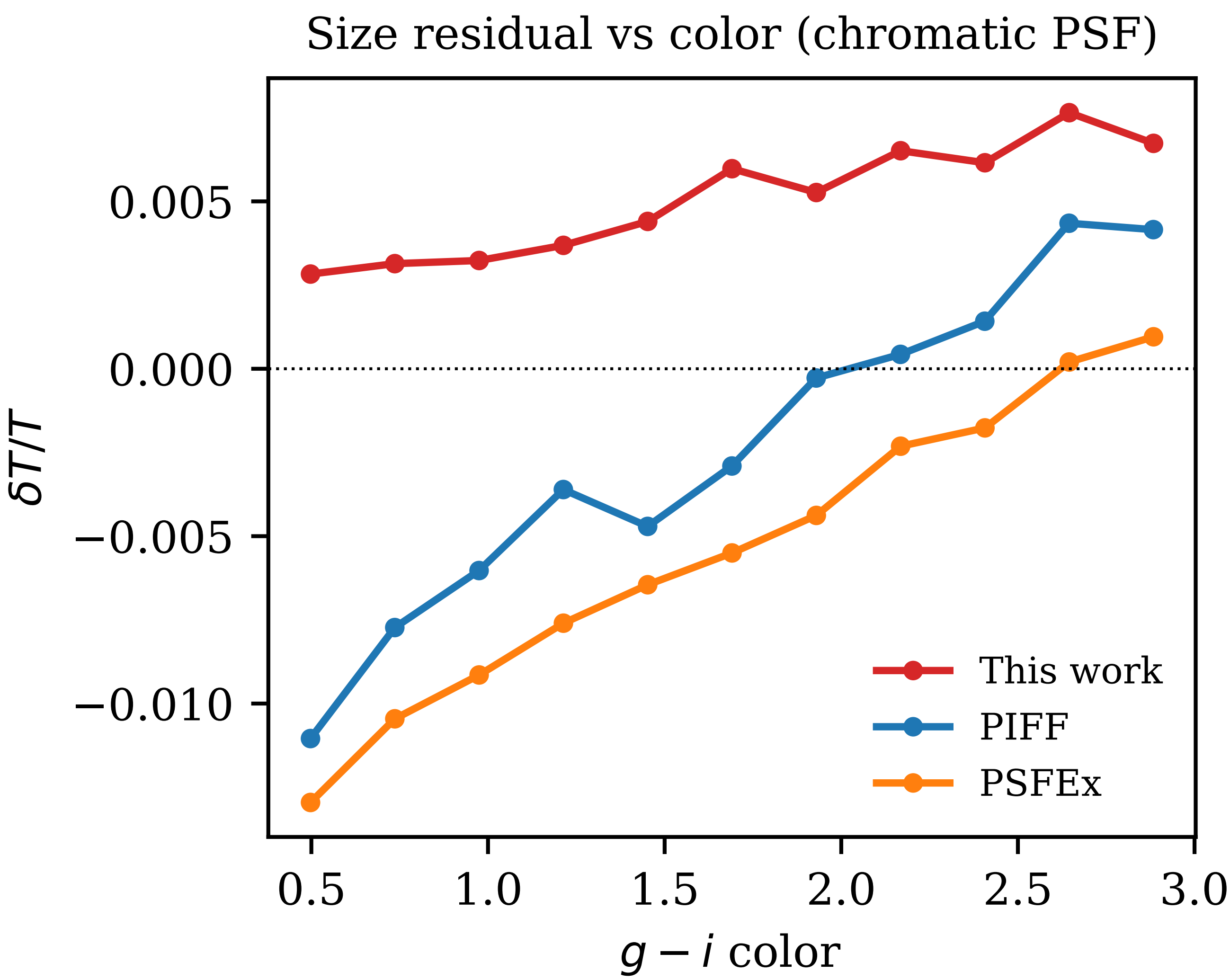

Figure 14. Size residual versus $g - i$ color (cf. PIFF Fig. 11); color conditioning keeps our chromatic residual flat.

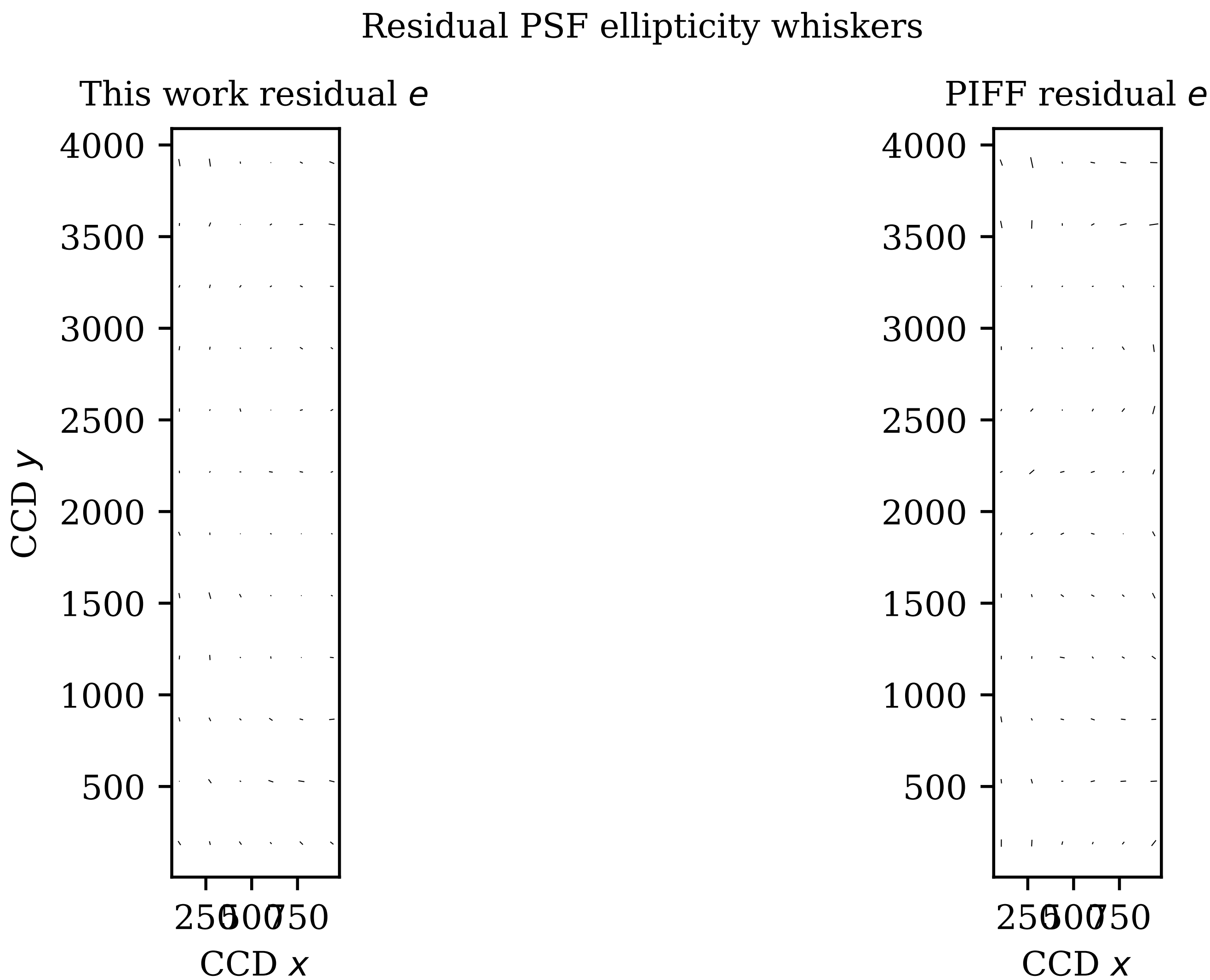

Figure 15. Residual PSF-ellipticity whiskers across the CCD, ImplicitPSF and PIFF (cf. PIFF Fig. 2).